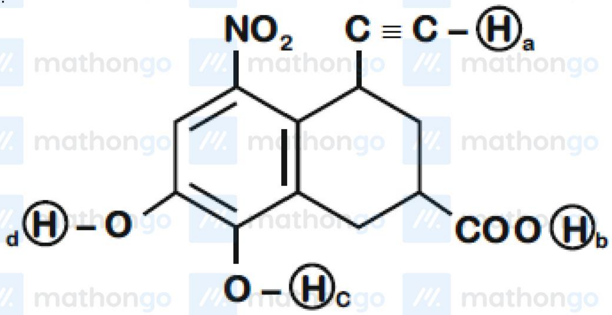
Q1 JEE Main 2020 - 9 January (Evening)

Which of the following has the shortest C-Cl bond?

(A) $Cl-CH=CH-NO_2$
(B) $Cl-CH=CH-CH_3$
(C) $Cl-CH=CH_2$
(D) $Cl-CH=CH-OCH_3$

Q2 JEE Main 2020 - 2 September (Evening)

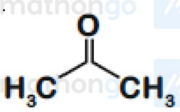
Arrange the following labelled hydrogens in decreasing order of acidity



(A) $b>c>d>a$
(B) $\mathrm{b}>\mathrm{a}>\mathrm{c}>\mathrm{d}$
(C) $c>b>d>a$
(D) $c>b>a>d$

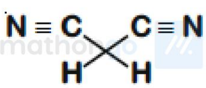
Q3 JEE Main 2020 - 3 September (Morning)

Which one of the following compounds possesses the most acidic hydrogen?

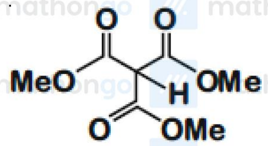


(A)

(B) $\mathrm{H_3C}-\mathrm{C}\equiv\mathrm{C}-\mathrm{H}$



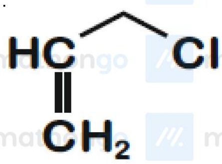
(C)



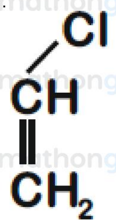
(D)

Q4 JEE Main 2020 - 4 September (Evening)

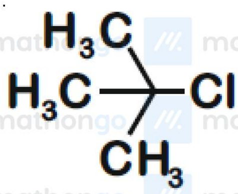
Among the following compounds, which one has the shortest $\mathrm{C}-\mathrm{Cl}$ bond?



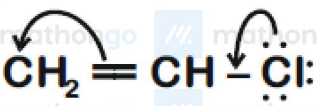
(A)



(B)



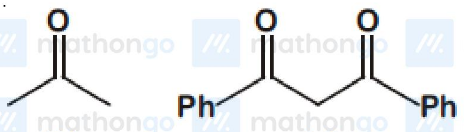
(C)



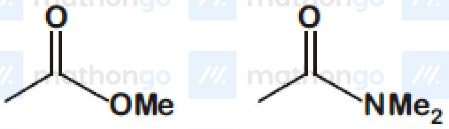
(D)

Q5 JEE Main 2020 - 5 September (Morning)

The increasing order of the acidity of the α -hydrogen of the following compounds is



(A) (B)

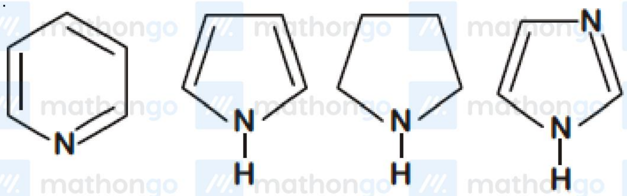


(C) (D)

(A) $(\mathrm{C})<(\mathrm{A})<(\mathrm{B})<(\mathrm{D})$
(B) $(B)<(C)<(A)<(D)$
(C) $(A)<(C)<(D)<(B)$
(D) $(\mathrm{D})<(\mathrm{C})<(\mathrm{A})<(\mathrm{B})$

Q6 JEE Main 2020 - 5 September (Morning)

The increasing order of basicity of the following compounds is

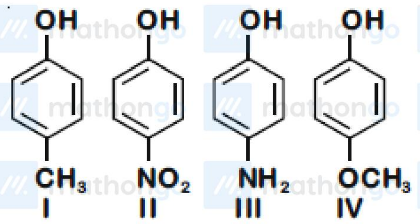


(A) (B) (C) (D)

(A) $(B)<(A)<(D)<(C)$
(B) $(\mathrm{D})<(\mathrm{A})<(\mathrm{B})<(\mathrm{C})$
(C) $(B)<(A)<(C)<(D)$
(D) $(\mathrm{A})<(\mathrm{B})<(\mathrm{C})<(\mathrm{D})$

Q7 JEE Main 2020 - 5 September (Evening)

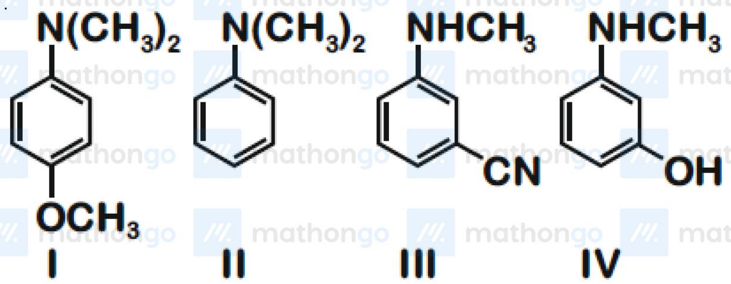
The increasing order of boiling points of the following compounds is



(A) $III < I < II < IV$
(B) $IV < I < II < III$
(C) $I < III < IV < II$
(D) $I < IV < III < II$

Q8 JEE Main 2020 - 6 September (Morning)

The increasing order of $\mathrm{pK_b}$ values of the following compounds is



(A) I < II < IV < III
(B) I < II < III < IV
(C) II < I < III < IV
(D) II < IV < III < I

Q9 JEE Main 2020 - 7 January (Morning)

The number of chiral centers in chloramphenicol is :

Q10 JEE Main 2020 - 7 January (Morning)

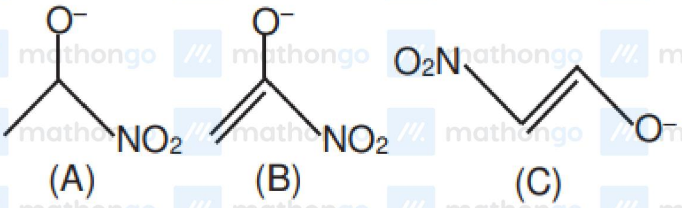
The dipole moments of $CHCl_3,\ CCl_4$ and CH_4 are in the order :

(A) $CHCl_3 > CCl_4 = CH_4$

(B) $CHCl_3 > CCl_4 > CH_4$
(C) $CCl_4 > CHCl_3 > CH_4$
(D) $CCl_4 = CH_4 > CHCl_3$

Q11 JEE Main 2020 - 7 January (Evening)

Stability order of following alkoxide ions is



(A) $C > B > A$
(B) $A > C > B$
(C) $B > A > C$
(D) $C > A > B$

Q12 JEE Main 2020 - 8 January (Morning)

Arrange the following compounds in increasing order of C-OH bond length

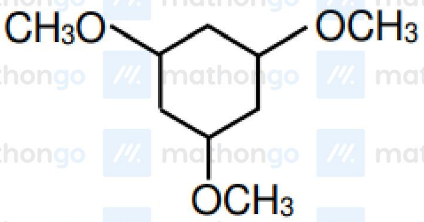
$$\underset{(A)}{\text{Methanol}} \quad \underset{(B)}{\text{Phenol}} \quad \underset{(C)}{p\text{-Ethoxyphenol}}$$

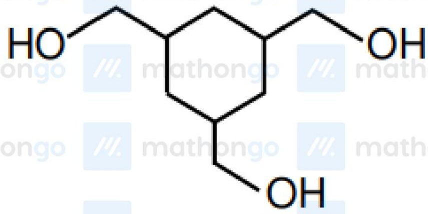
(A) $A > B > C$
(B) $A > C > B$
(C) $C > B > A$
(D) $B > C > A$

Q13 JEE Main 2020 - 8 January (Evening)

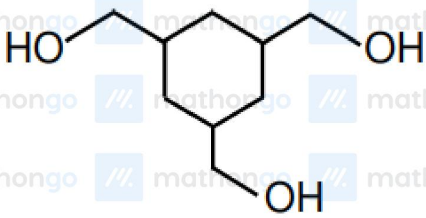
Among the compound A and B with molecular formula $C_9H_{18}O_3$, A is having higher boiling point than B. The possible structures of A and B are :

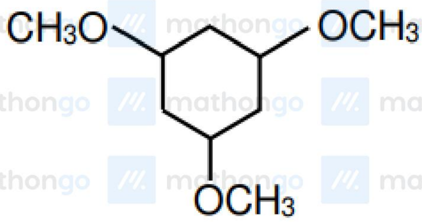
(A)

A = 

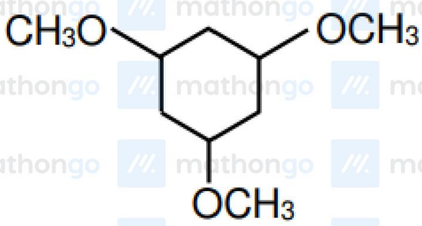
B = 

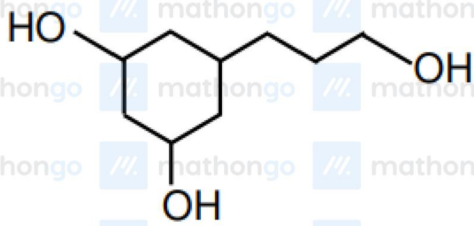
(B)

A = 

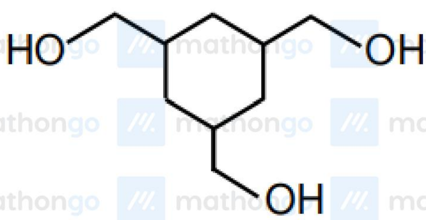
B = 

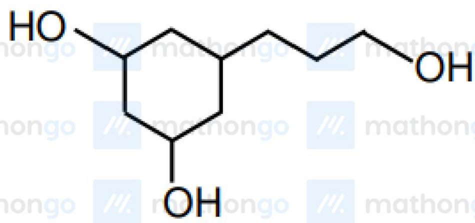
(C)

A = 

B = 

(D)

A = 

B = 

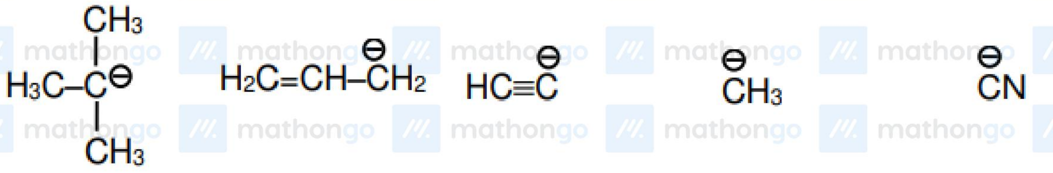
Q14 JEE Main 2020 - 9 January (Morning)

A chemist has 4 samples of artificial sweetener $A,\ B,\ C$ and D. To identify these samples, he performed certain experiments and noted the following observations :
(i) A and D both form blue-violet colour with ninhydrin.
(ii) Lassaigne extract of C gives positive $AgNO_3$ test and negative $Fe_4\left[Fe(CN)_6\right]_3$ test.
(iii) Lassaigne extract of B and D gives positive sodium nitroprusside test.
Based on these observations which option is correct ?

(A) A - Saccharine, B - Aspartame, C - Sucralose, D - Alitame
(B) A - Aspartame, B - Saccharine, C - Sucralose, D - Alitame
(C) A - Saccharine, B - Aspartame, C - Alitame, D - Sucralose
(D) A - Aspartame, B - Sucralose, C - Saccharine, D - Alitame

Q15 JEE Main 2020 - 9 January (Morning)

The increasing order of basicity for the following intermediates is (from weak to strong)

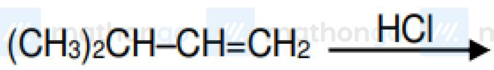


(i) (ii) (iii) (iv) (v)

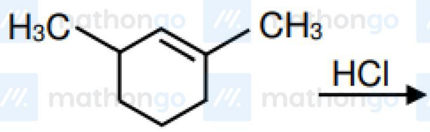
(A) $(iii) < (iv) < (ii) < (i) < (v)$
(B) $(iii) < (i) < (ii) < (iv) < (v)$
(C) $(v) < (iii) < (ii) < (iv) < (i)$
(D) $(v) < (i) < (iv) < (ii) < (iii)$

Q16 JEE Main 2020 - 9 January (Evening)
Which of the following reactions will not produce a racemic product?

(A)



(B)



(C)

$$CH_3CH_2\text{–}CH=CH_2 \xrightarrow{HBr}$$

(D)

$$CH_3CH_2\text{–}\underset{\underset{\textstyle O}{\|}}{C}\text{–}CH_3 \xrightarrow{HCN}$$

Answer Key

Q1 (A)	**Q2** (A)	**Q3** (D)	**Q4** (B)
Q5 (D)	**Q6** (C)	**Q7** (None)	**Q8** (A)
Q9 (2)	**Q10** (A)	**Q11** (A)	**Q12** (B)
Q13 (B)	**Q14** (B)	**Q15** (C)	**Q16** (A)

Q1

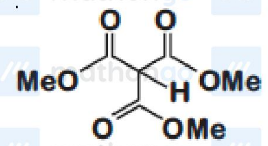
Resonance form of $Cl - CH = CH - NO_2$ is more stable than resonance form of any other given compounds. Hence, double bond character in carbon-chlorine bond is maximum and bond length is shortest.

Q2

$b > c > d > a$

Q3

Compound with most acidic hydrogen among the given compounds is

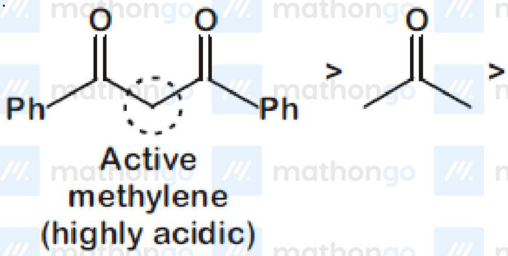


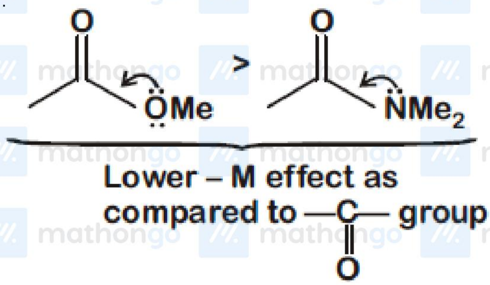
Q4

In

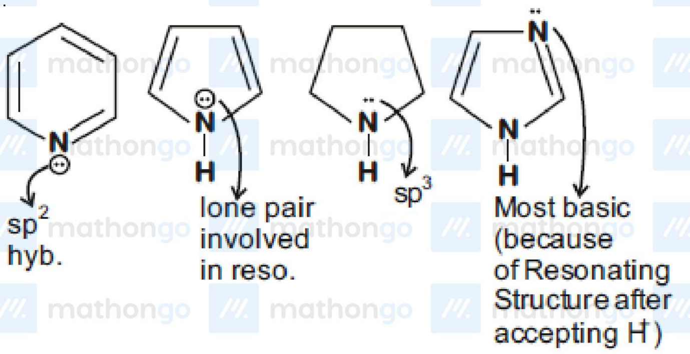
due to resonance, $\mathrm{C} - \mathrm{Cl}$ bond acquires partial double bond character and has shortest bond length among given species.

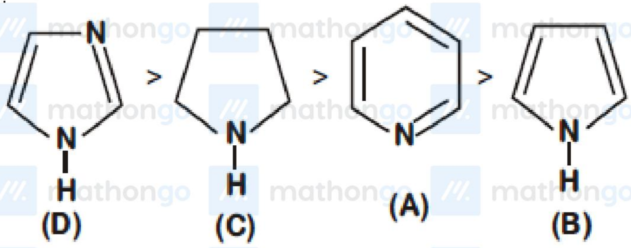
Q5


Ph
Ph
Active
methylene
(highly acidic)


OMe
NMe2
Lower – M effect as
compared to —C— group

Q6

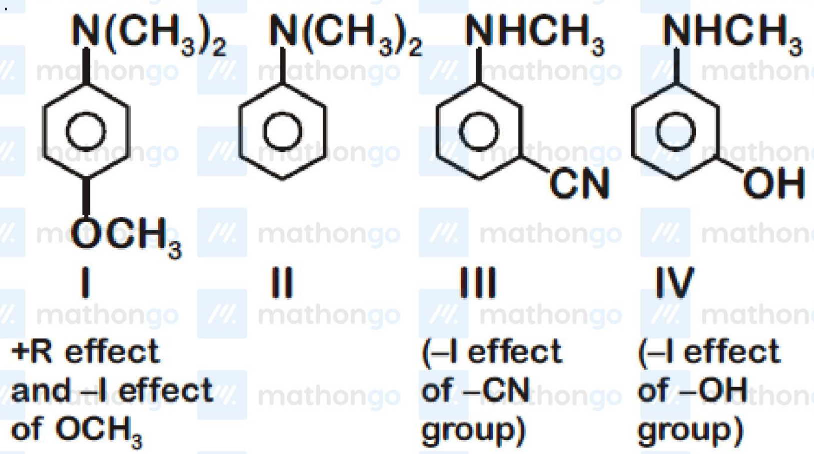

sp2
hyb.
lone pair
involved
in reso.
sp3
Most basic
(because
of Resonating
Structure after
accepting H+)


(D)
(C)
(A)
(B)

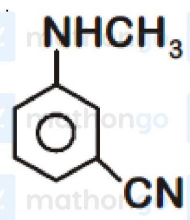
Q7

Order of boiling point of the following compound is $I < IV < II < III$
$I \rightarrow$ B.P. $202^{\circ}\mathbf{C}$
$II \rightarrow$ B.P. $279^{\circ}\mathrm{C}$
$III \rightarrow$ B.P. $284^{\circ}\mathrm{C}$
$\mathrm{IV} \rightarrow \mathrm{B.P.} \quad 243^{\circ}\mathrm{C}$

Q8



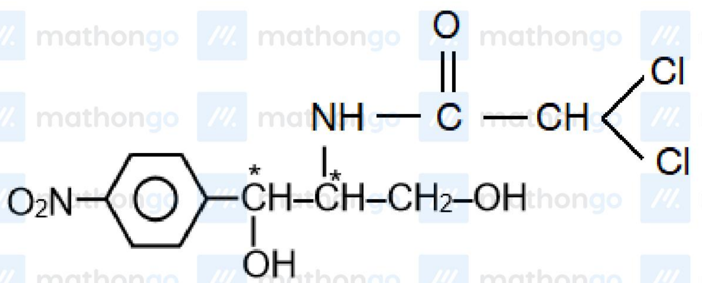
$-\mathrm{OCH}_3$ group increases electron density of ring at O and P position making (I) most basic.



is least basic due to -1 effect of $-\mathrm{CN}$
group at meta position.
since -1 effect of $-\mathrm{CN} > -1$ effect of $-\mathrm{OH}$
group

Hence correct basic strength will follow the order
I>II>IV>III
Basic strength $\propto \frac{1}{\mathrm{pK_b\ value}}$
Order of $\mathrm{K_b}$ value
I < II < IV < III

Q9



Q10

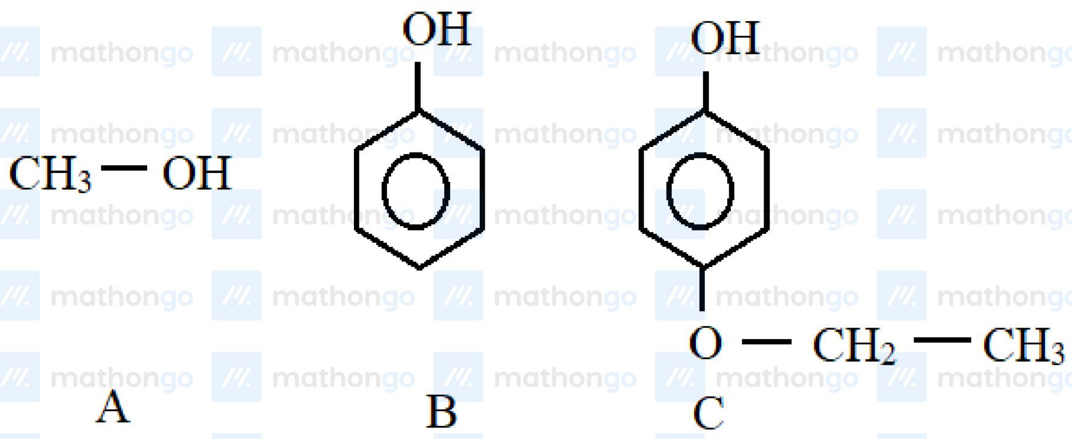
$\mu_{CCl_4} = \mu_{CH_4} = 0$ but $\mu_{CHCl_3} \neq 0$

Q11

When negative charge is delocalised with electron withdrawing group like (NO_2) then stability increases.
(A) Negative charge is delocalised with NO_2 group
(B) Negative charge is delocalised with carbon of alkene
(C) Negative charge is localised

Q12

There is not any resonance in $CH_3 - OH$. Resonance is poor in p-Ethoxyphenol than phenol.so $C - OH$ bond length order is $A > C > B$



Q13

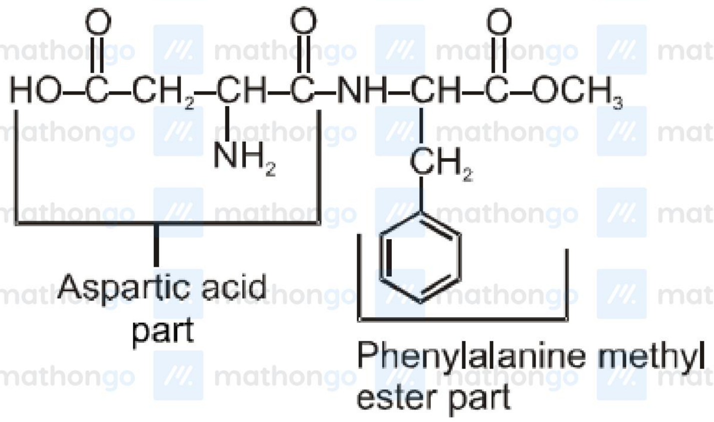
In (A), extensive inter-molecular H-bonding is possible while in (B) there is no Inter-molecular H-bonding.

Q14

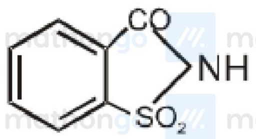
(i) A & D give positive test with ninhydrin because both have free carboxylic and amine groups.
(ii) C form precipitate with $AgNO_3$ in the lassaigne extract of the sugar because it has chlorine atoms.

(iii) B & D give positive test with sodium nitroprusside because both have sulphur atoms.

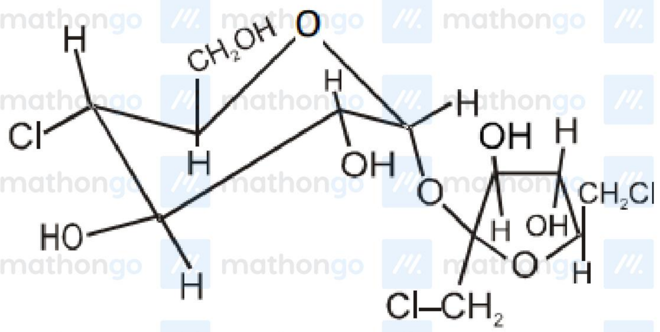
A – Aspartame



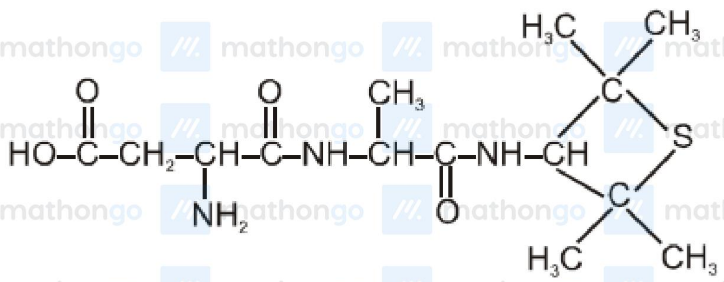
B – Saccharine



C – Sucralose



D – Alitame



Basicity is inversely proportional to electronegativity.

Q16

In this reaction, major product is not chiral.